Other Services

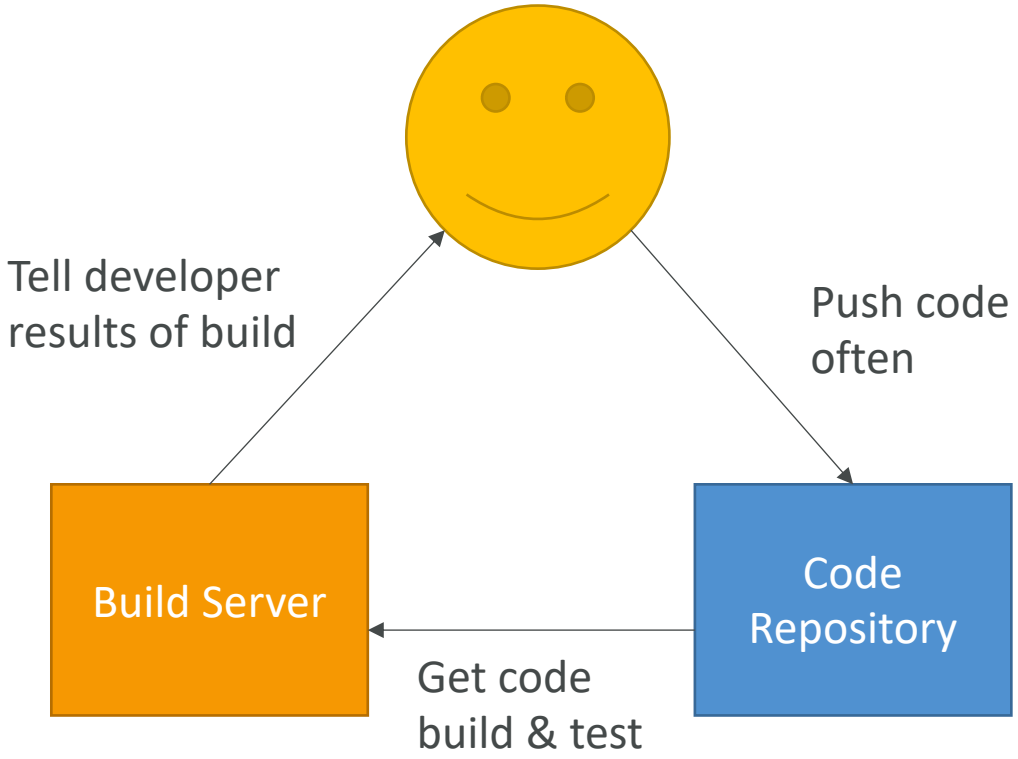
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Continuous Integration

- Developers push the code to a code repository often (GitHub / CodeCommit / Bitbucket / etc…)
- A testing / build server checks the code as soon as it's pushed (CodeBuild / Jenkins CI / etc…)
- The developer gets feedback about the tests and checks that have passed / failed

- Find bugs early, fix bugs
- Deliver faster as the code is tested
- Deploy often
- Happier developers, as they're unblocked

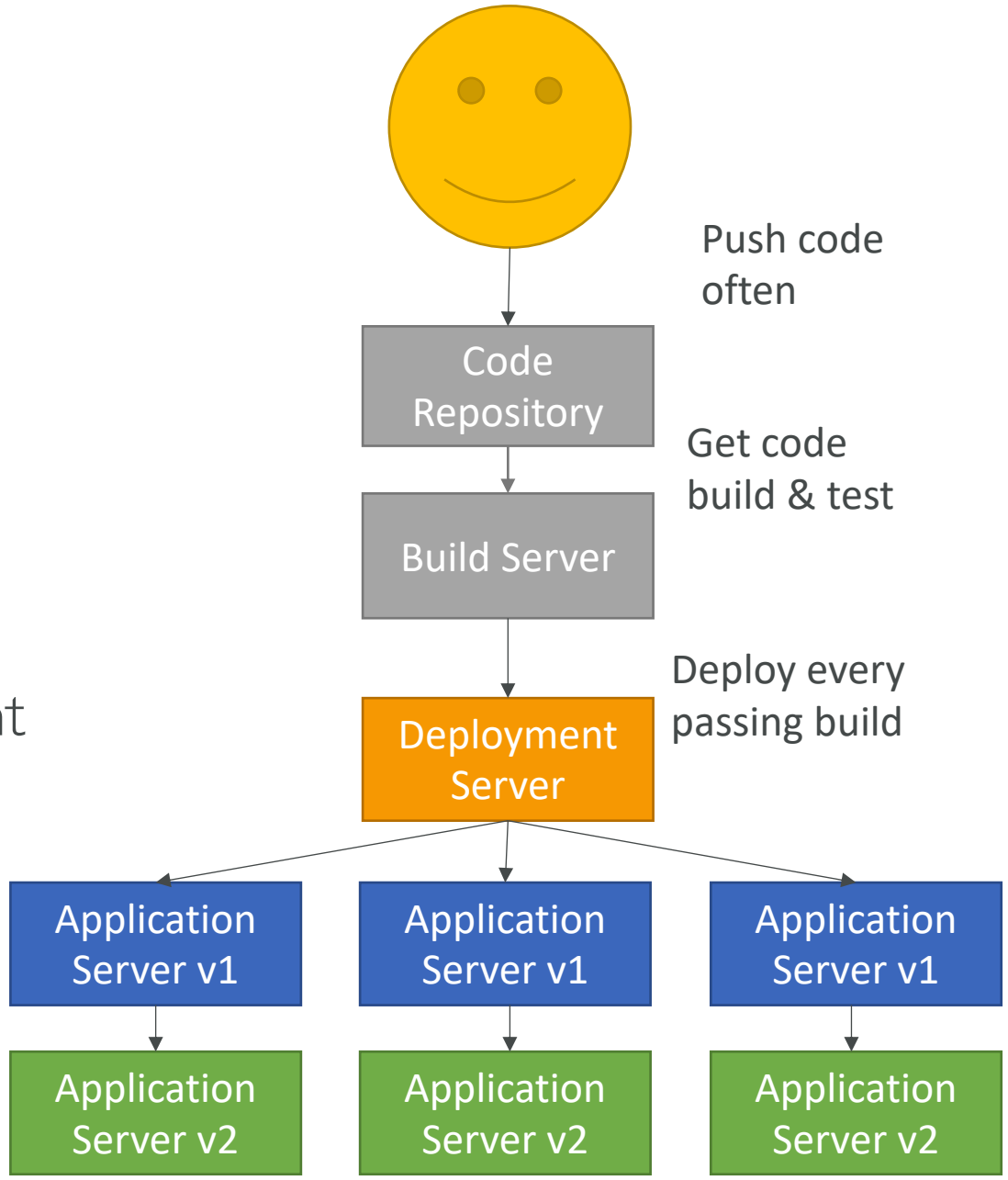


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Continuous Delivery

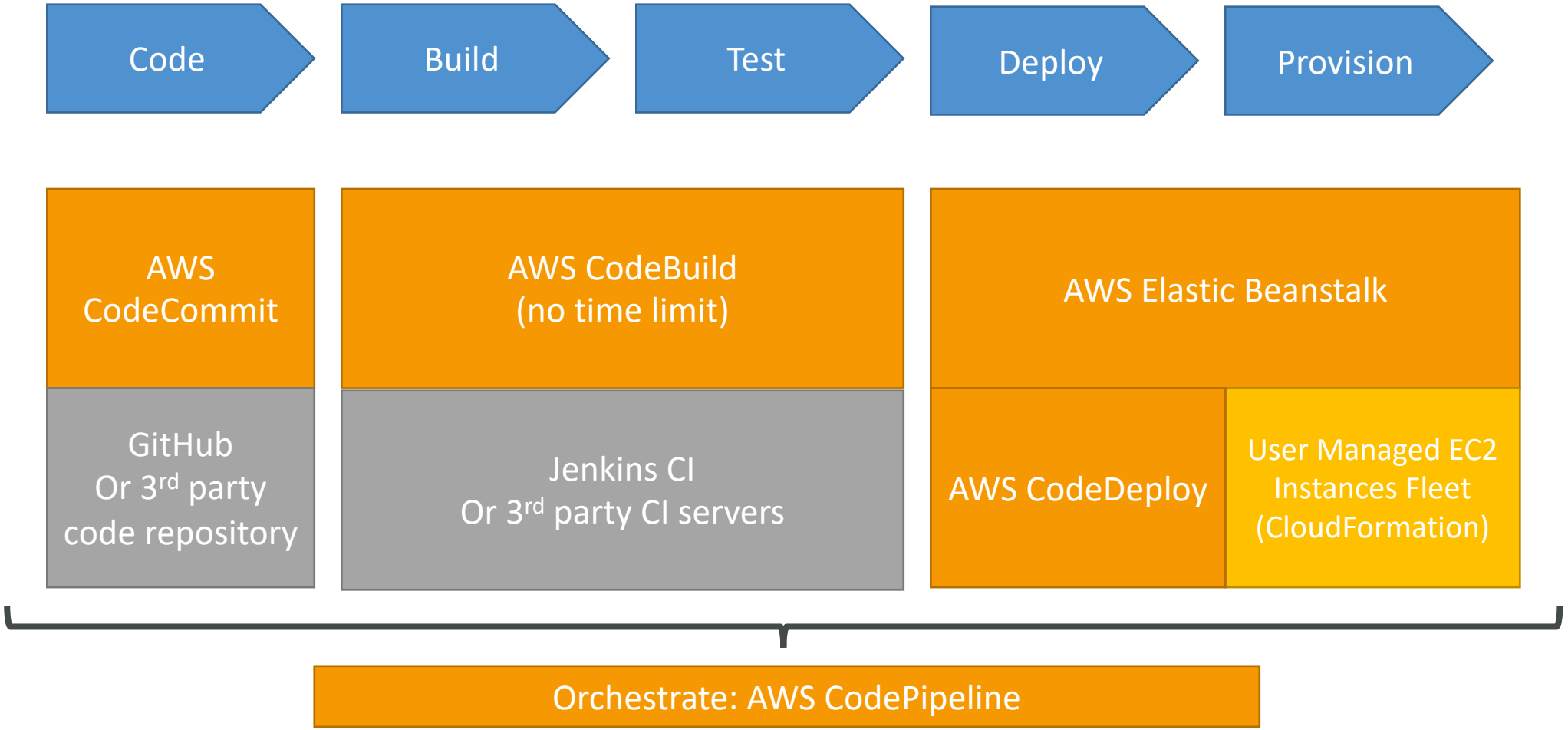
- Ensure that the software can be released reliably whenever needed.
- Ensures deployments happen often and are quick
- Shift away from "one release every 3 months" to "5 releases a day"
- That usually means automated deployment
  - CodeDeploy
  - Jenkins CD
  - Spinnaker
  - Etc…



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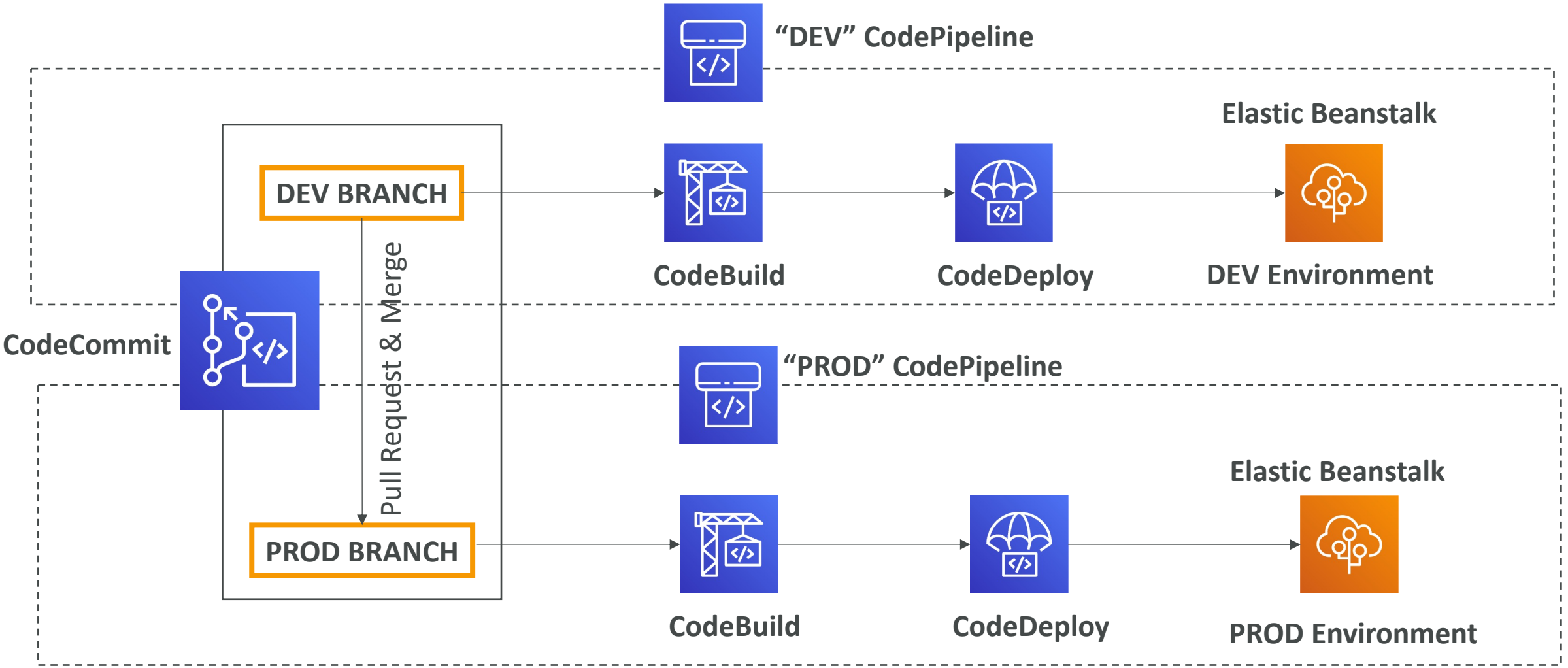
Technology Stack for CICD



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CICD Architecture

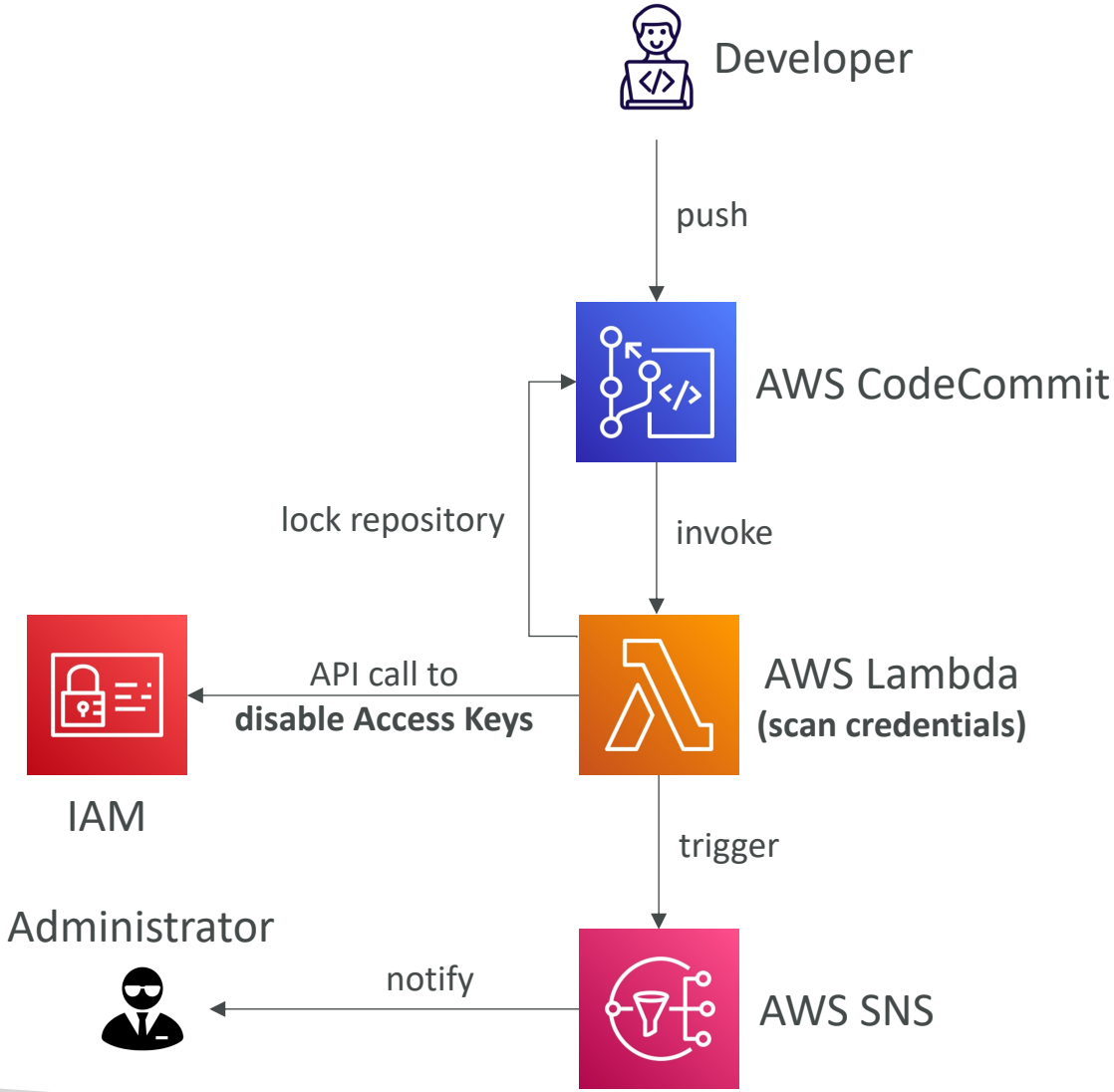


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CodeCommit Trigger for AWS Lambda

- Every push to CodeCommit can trigger a Lambda function
- The Lambda function can scan for leaked AWS credentials on every code push, and disable them automatically to remedy the issue



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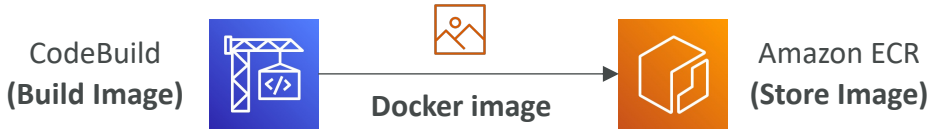
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Good to know – CICD

- You can use a **manual approval stage** in CodePipeline
- Running unit tests CodeCommit + CodeBuild + Code Pipeline



- Build and Store Docker Images: CodeBuild + ECR

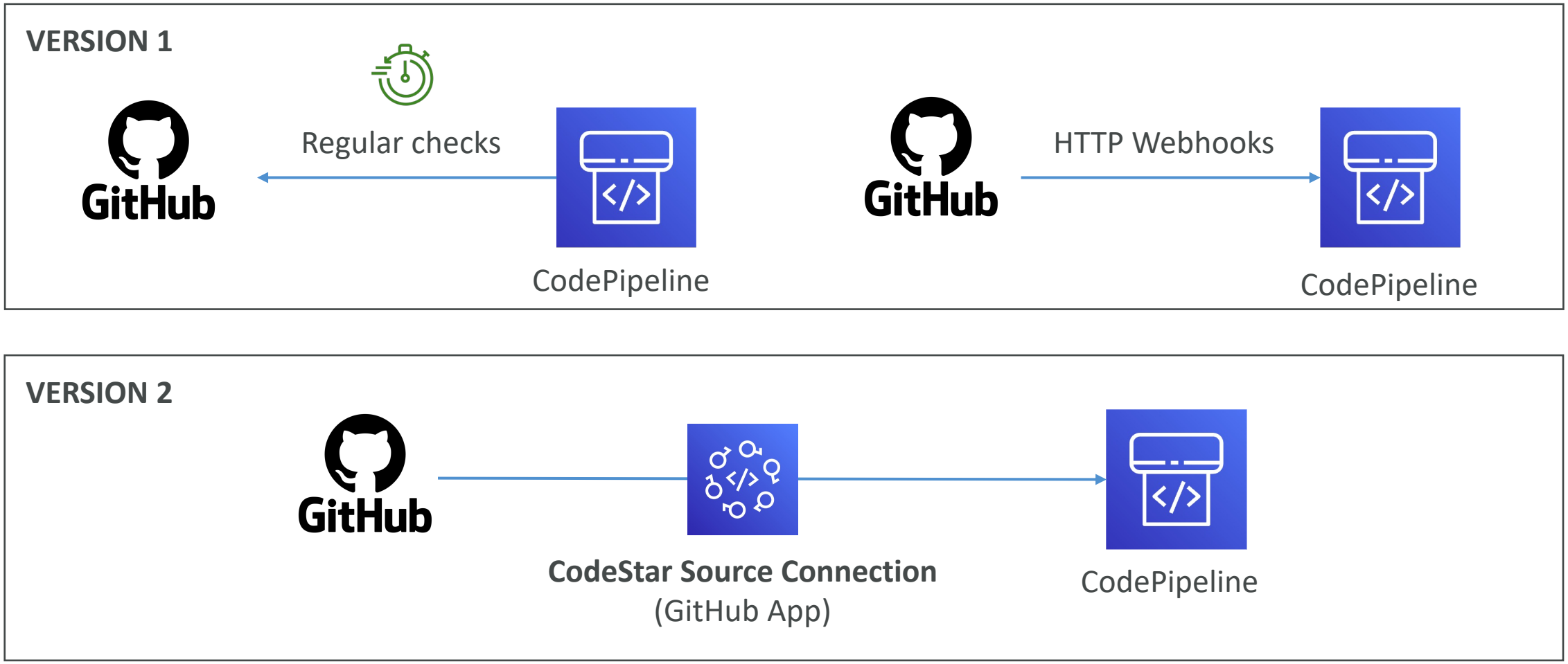


- Automated CloudFormation deployment: CodePipeline

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CodePipeline – GitHub integration

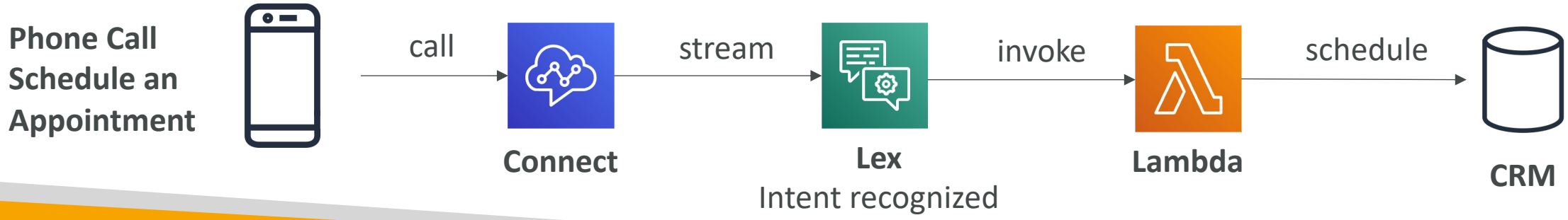


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Alexa for Business, Lex & Connect

- Alexa for Business:
  - Use Alexa to help employees be more productive in meeting rooms and their desk
  - Measure and increase the utilization of meeting rooms in their workplace
- Amazon Lex: (same technology that powers Alexa)
  - Automatic Speech Recognition (ASR) to convert speech to text
  - Natural Language Understanding to recognize the intent of text, callers
  - Helps build chatbots, call center bots
- Amazon Connect:
  - Receive calls, create contact flows, cloud-based virtual contact center
  - Can integrate with other CRM systems or AWS



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Kinesis Video Streams

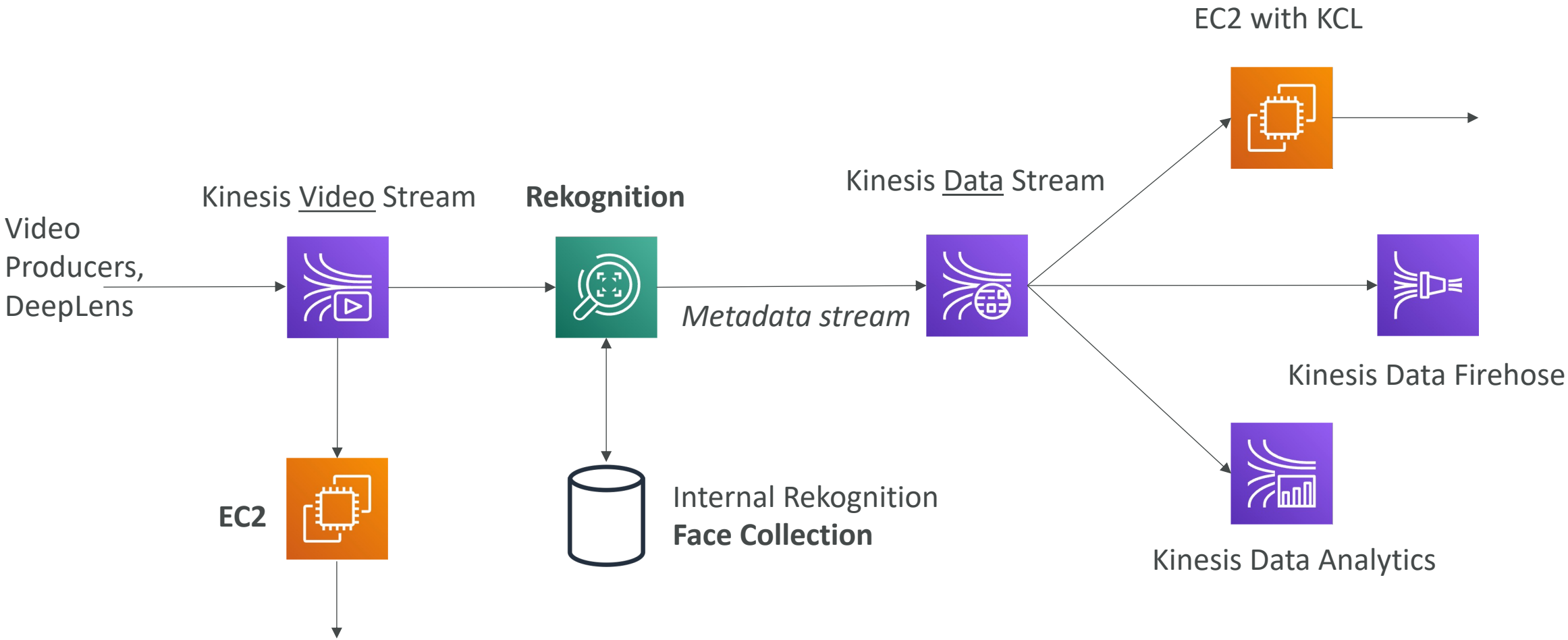


- One video stream per streaming device (producers)
  - Security cameras, body worn camera, smartphone
  - Can use a Kinesis Video Streams Producer library
- Underlying data is stored in S3 (but we don't have access to it)
- Cannot output the stream data to S3 (must build custom solution)
- Consumers:
  - Consumed by EC2 instances for real time analysis, or in batch
  - Can leverage the Kinesis Video Stream Parser Library
  - Integration with AWS Rekognition for facial detection

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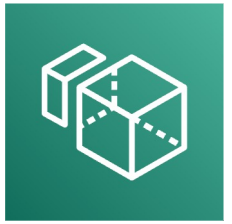
Video Streaming & Rekognition



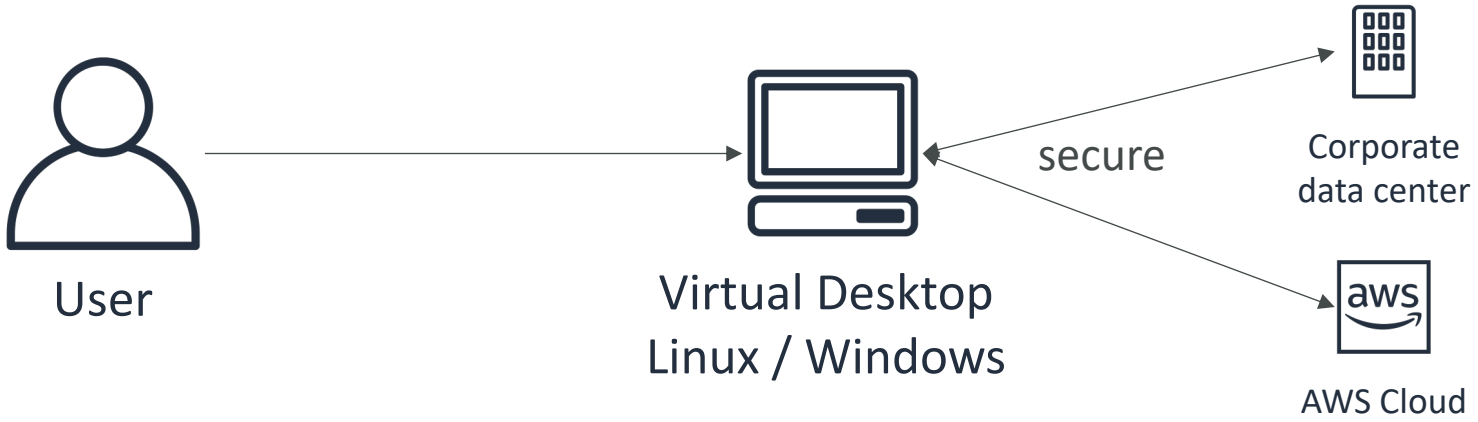
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Amazon WorkSpaces



- Managed, Secure Cloud Desktop
- Great to eliminate management of on-premises VDI (Virtual Desktop Infrastructure)
- Pricing is either on-demand (pay per hour) or monthly subscription
- Secure, Encrypted, Network Isolation
- Integrated with Microsoft Active Directory



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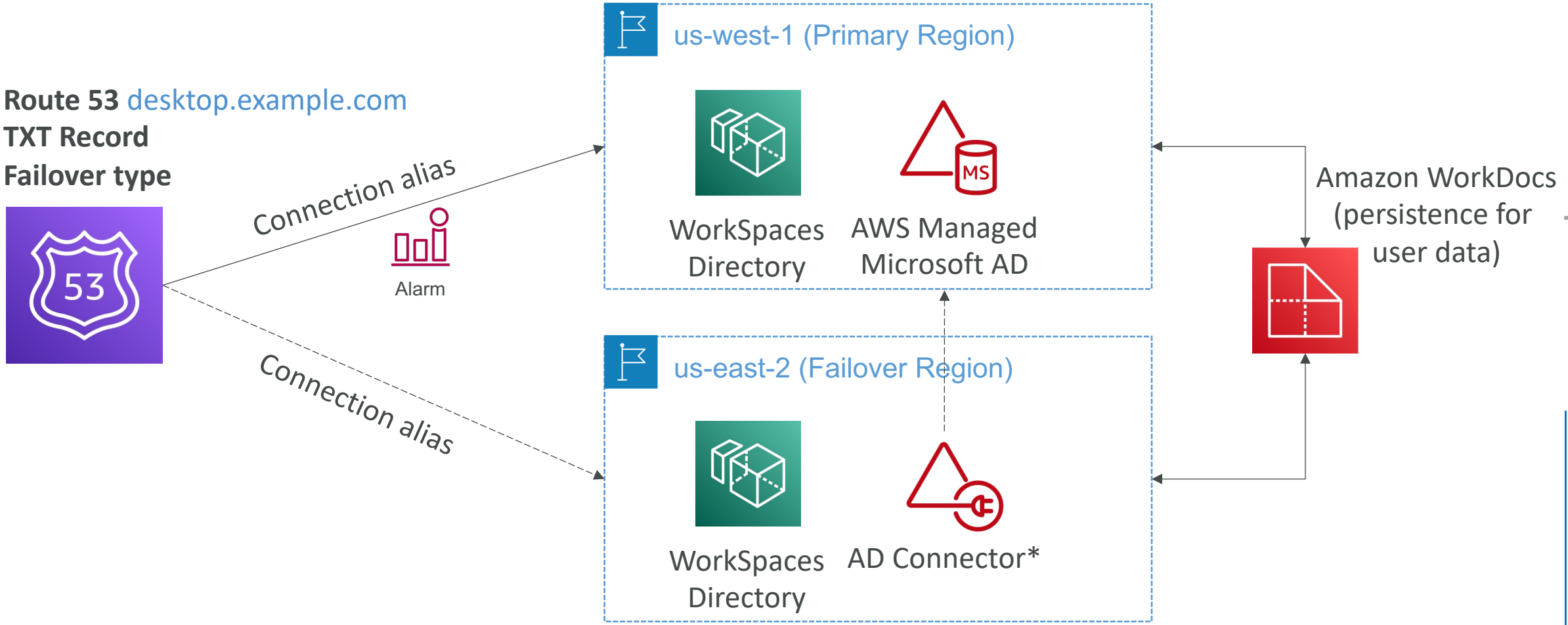
Amazon WorkSpaces

- WorkSpaces Application Manager (WAM)
  - Deploy and Manage applications as virtualized application containers
  - Provision at scale, and keep the applications updated using WAM
- Windows Updates
  - By default, Amazon Workspaces are configured to install software updates
  - Amazon WorkSpaces with Windows will have Windows Update turned on
  - You have full control over the Windows Update frequency
- Maintenance Windows
  - Updates are installed during maintenance windows (you define them)
  - Always On WorkSpaces: default is from 00h00 to 04h00 on Sunday morning
  - AutoStop WorkSpaces: automatically starts once a month to install updates
  - Manual maintenance: you define your windows and perform maintenance

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Amazon WorkSpaces - Cross Region Redirection



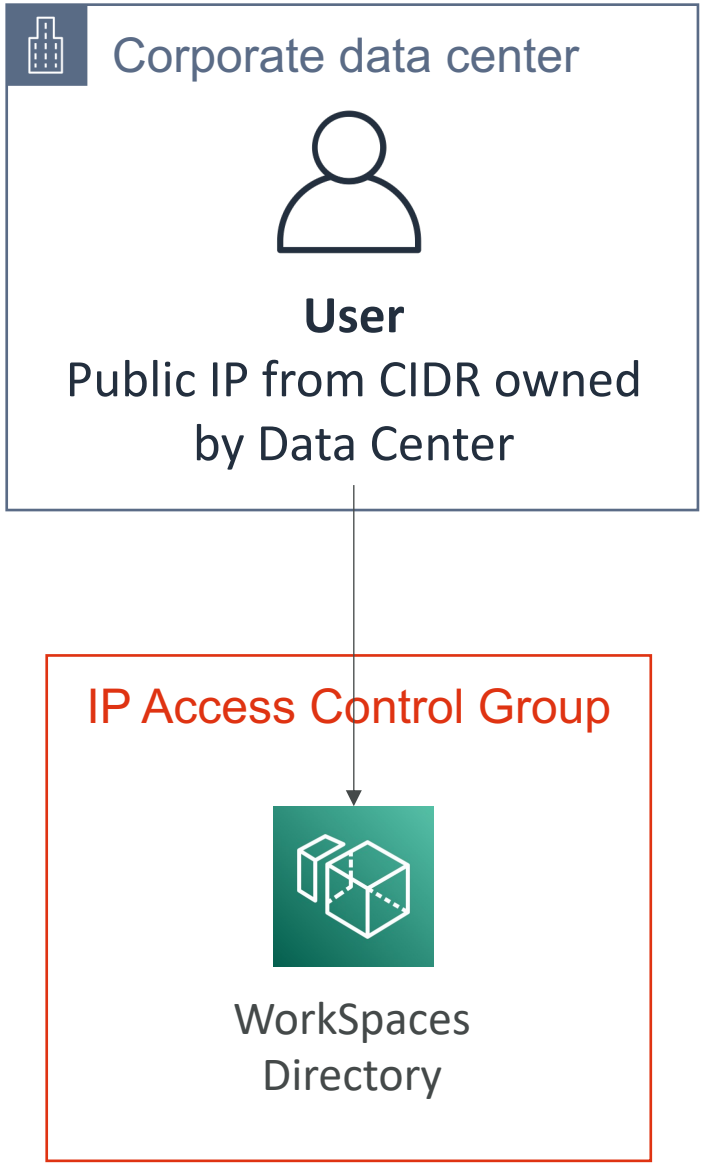
*must use AD Connector, can't use multi-region AWS Managed Microsoft AD

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Amazon WorkSpaces IP Access Control Groups

- Similar to security groups for Amazon WorkSpaces
- List of IP addresses / CIDR address ranges that users are authorized to connect from
- If users access WorkSpaces through VPN or NAT, the IP Access Control Group must authorize the public IP of these



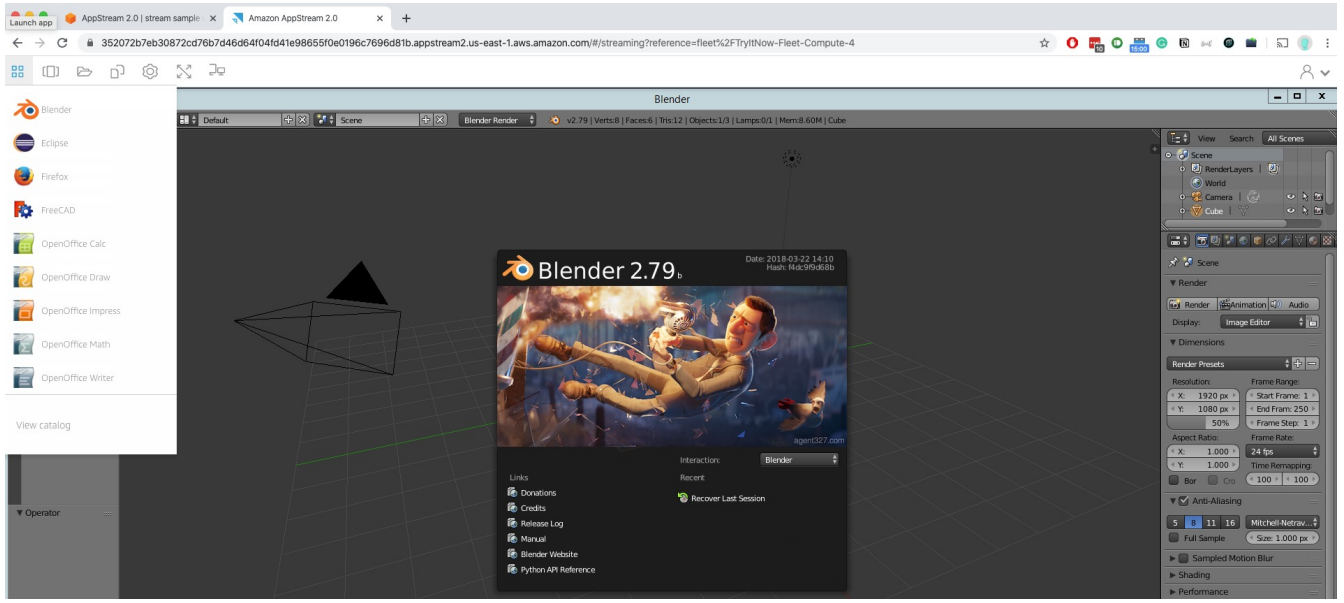
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Amazon AppStream 2.0



- Desktop Application Streaming Service
- Deliver to any computer, without acquiring, provisioning infrastructure
- The application is delivered from within a web browser



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Amazon AppStream 2.0 vs WorkSpaces

- Workspaces
  - Fully managed VDI and desktop available
  - The users connect to the VDI and open native or WAM applications
  - Workspaces are on-demand or always on

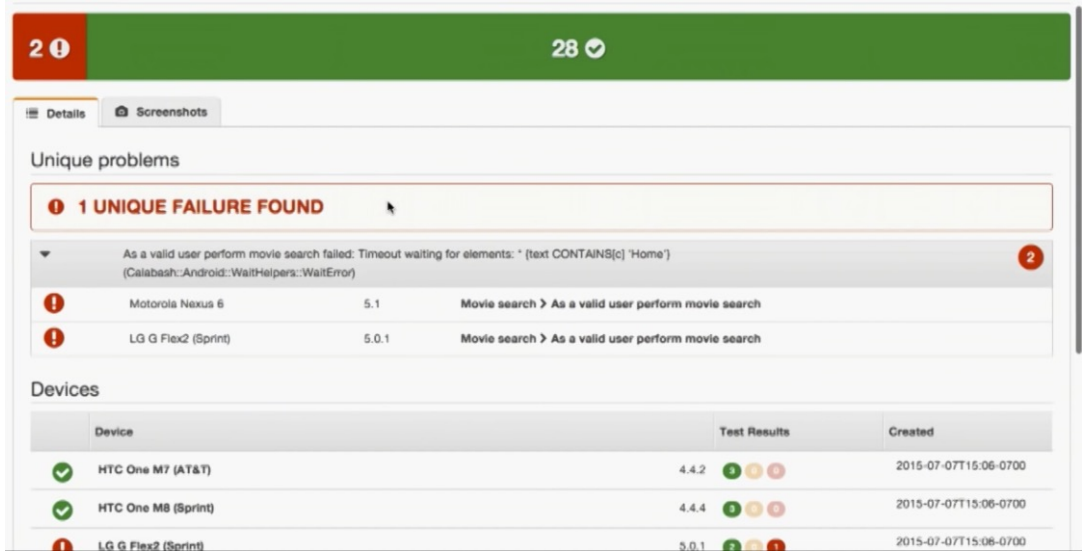
- AppStream 2.0
  - Stream a desktop application to web browsers (no need to connect to a VDI)
  - Works with any device (that has a web browser)
  - Allow to configure an instance type per application type (CPU, RAM, GPU)

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AWS Device Farm

- Application testing service for your mobile and web applications
- Test across **real browsers** and **real mobiles devices**
- **Fully automated** using framework
- Improve the quality of web and mobile apps
- Generates videos and logs to document the issues encountered
- Can **remotely log-in** to devices for debugging



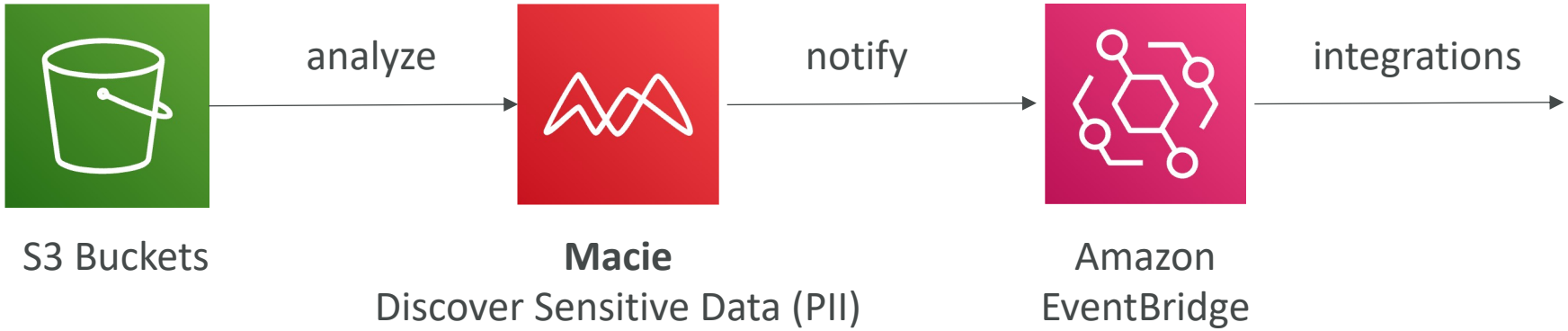
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AWS Macie



- Amazon Macie is a fully managed data security and data privacy service that uses **machine learning and pattern matching to discover and protect your sensitive data in AWS.**
- Macie helps identify and alert you to **sensitive data, such as personally identifiable information (PII)**



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Amazon Simple Email Service (Amazon SES)



- **Fully managed service to send emails securely, globally and at scale**
- Allows inbound/outbound emails
- Reputation dashboard, performance insights, anti-spam feedback
- Provides statistics such as email deliveries, bounces, feedback loop results, email open
- Supports DomainKeys Identified Mail (DKIM) and Sender Policy Framework (SPF)
- Flexible IP deployment: shared, dedicated, and customer-owned IPs
- Send emails using your application using AWS Console, APIs, or SMTP
- Use cases: transactional, marketing and bulk email communications

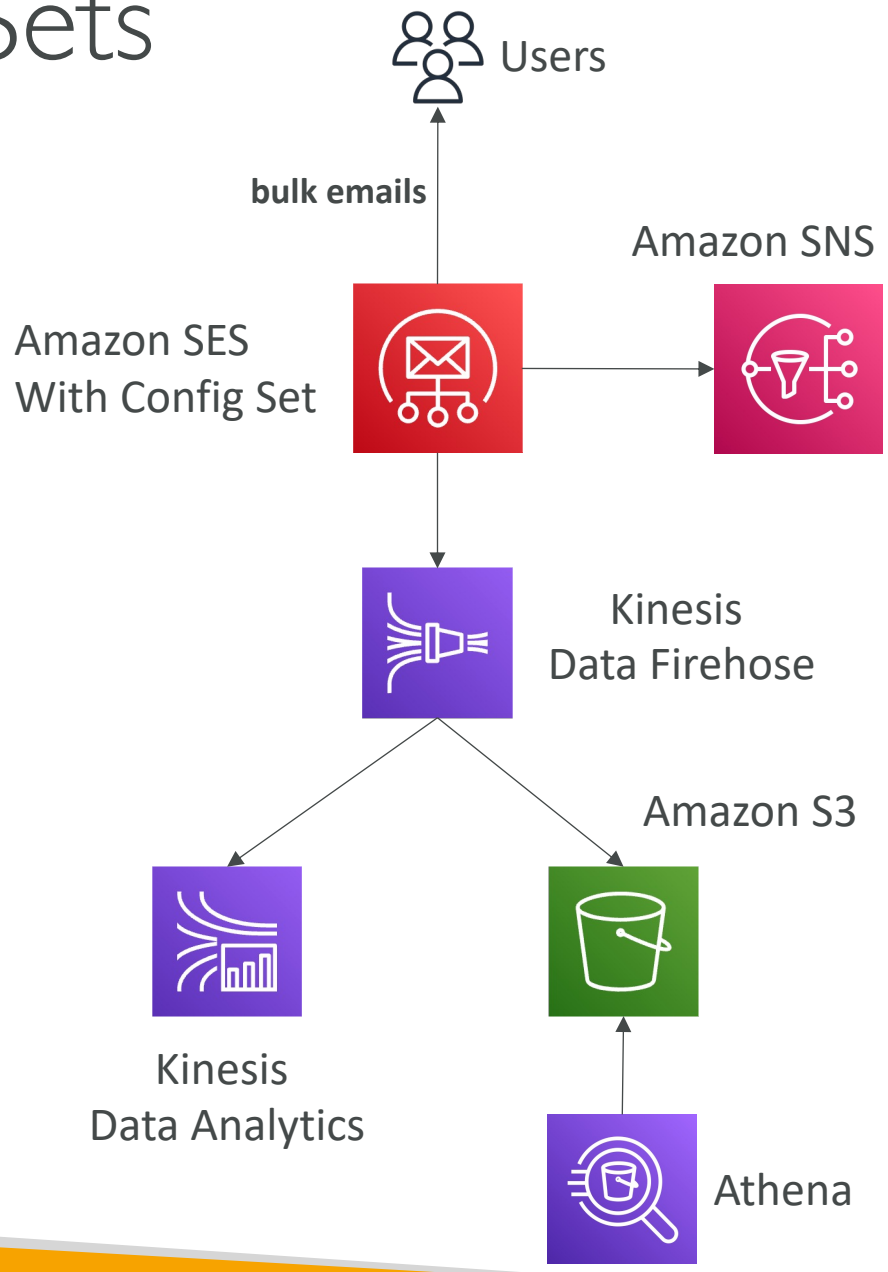


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Amazon SES – Configuration Sets

- Configuration sets help you customize and analyze your email send events
- **Event destinations:**
  - Kinesis Data Firehose: receives metrics (numbers of sends, deliveries, opens, clicks, bounces, and complaints) for each email
  - SNS: for immediate feedback on bounce and complaint information
- **IP pool management**: use IP pools to send particular types of emails

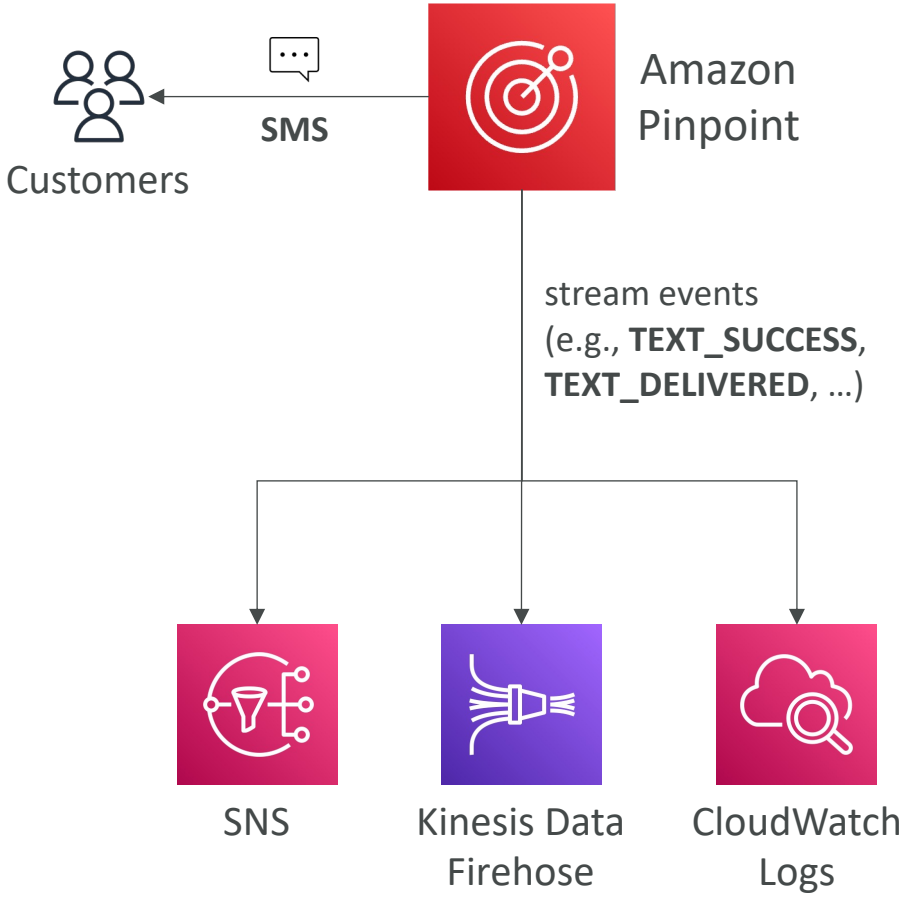


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Amazon Pinpoint

- Scalable **2-way (outbound/inbound)** marketing communications service
- Supports email, SMS, push, voice, and in-app messaging
- Ability to segment and personalize messages with the right content to customers
- Possibility to receive replies
- Scales to billions of messages per day
- Use cases: run campaigns by sending marketing, bulk, transactional SMS messages
- **Versus Amazon SNS or Amazon SES**
  - In SNS & SES you managed each message's audience, content, and delivery schedule
  - In Amazon Pinpoint, you create message templates, delivery schedules, highly-targeted segments, and full campaigns



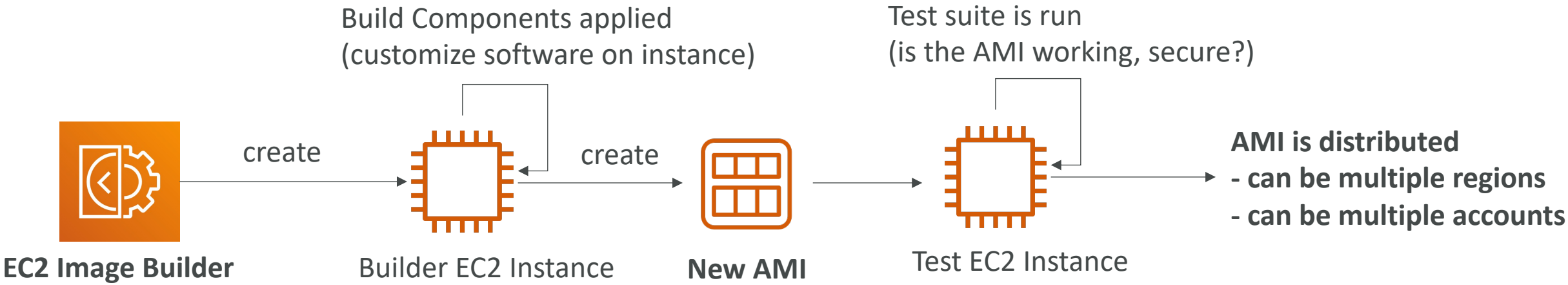
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EC2 Image Builder



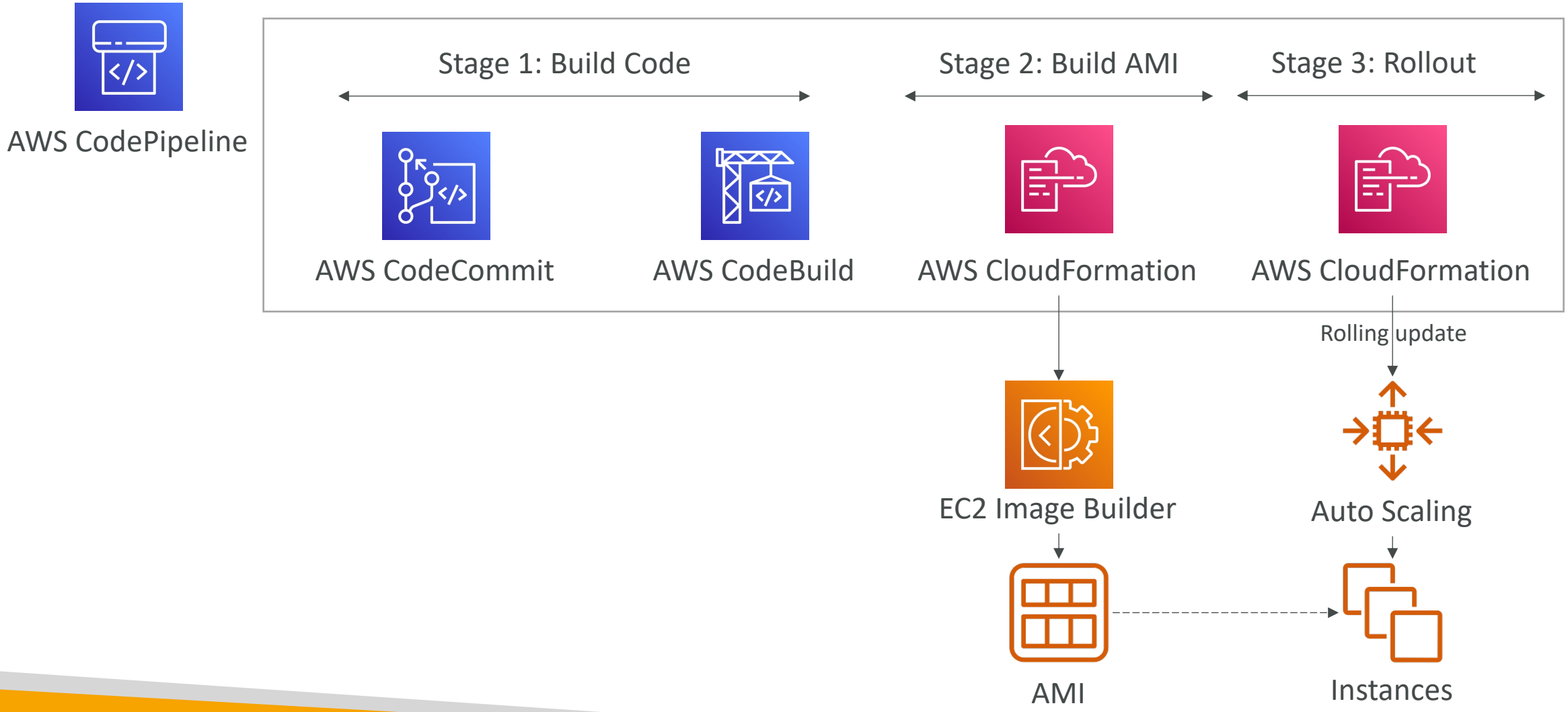
- Used to automate the creation of Virtual Machines or container images
- => Automate the creation, maintain, validate and test **EC2 AMIs**
- Can be run on a schedule (weekly, whenever packages are updated, etc…)
- Free service (only pay for the underlying resources)
- **Can publish AMI to multiple regions and multiple accounts**



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EC2 Image Builder – CICD Architecture



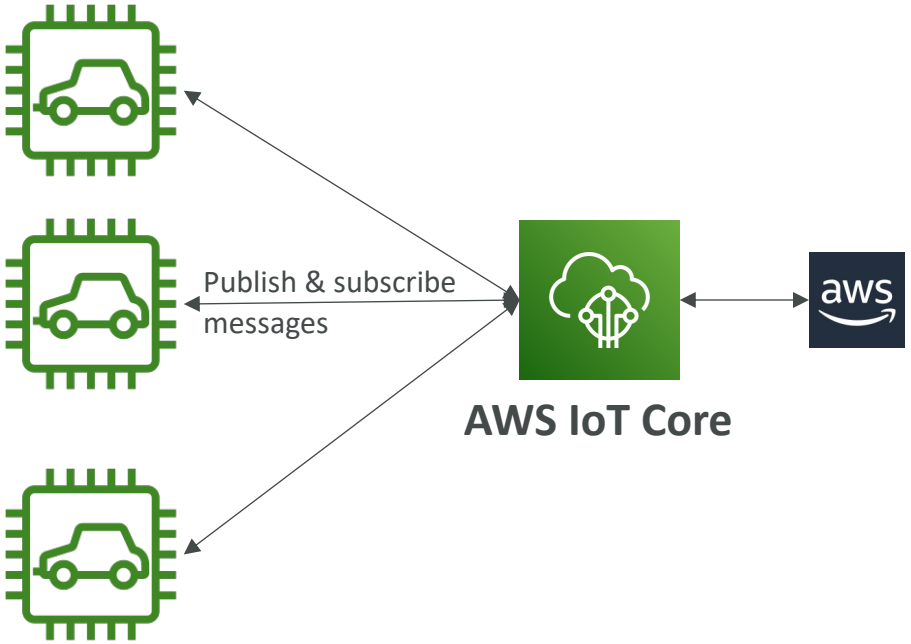
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AWS IoT Core



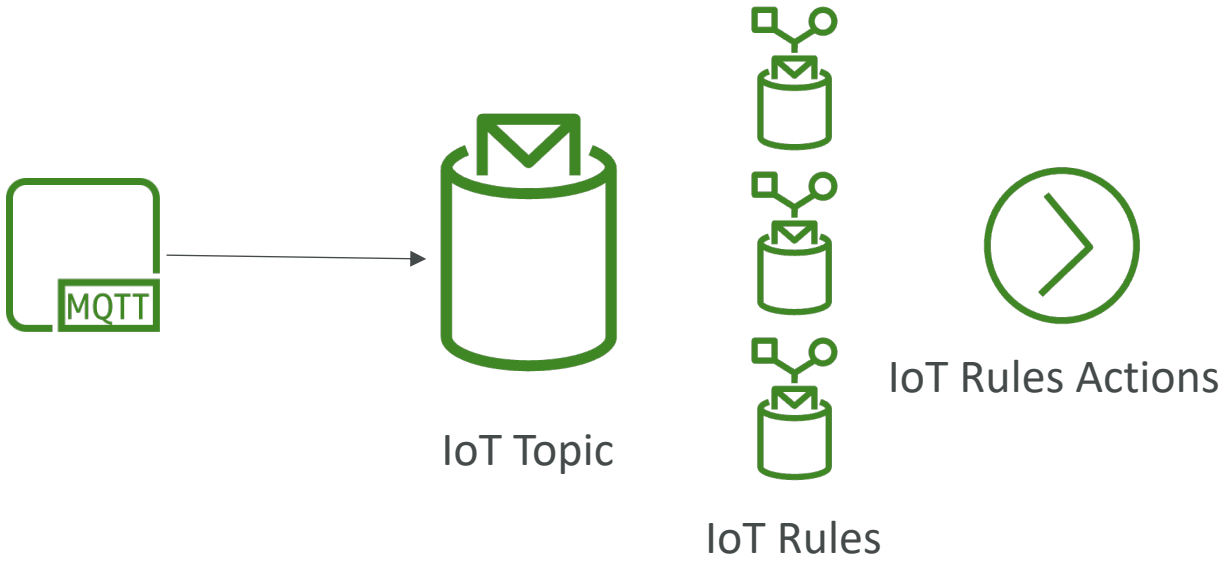
- IoT stands for "Internet of Things" – the network of internet-connected devices that are able to collect and transfer data
- AWS IoT Core allows you to **easily connect IoT devices to the AWS Cloud**
- **Serverless, secure & scalable** to billions of devices and trillions of messages
- Integrates with a lot of AWS services (Lambda, S3, SageMaker, etc.)
- Build IoT applications that gather, process, analyze, and act on data

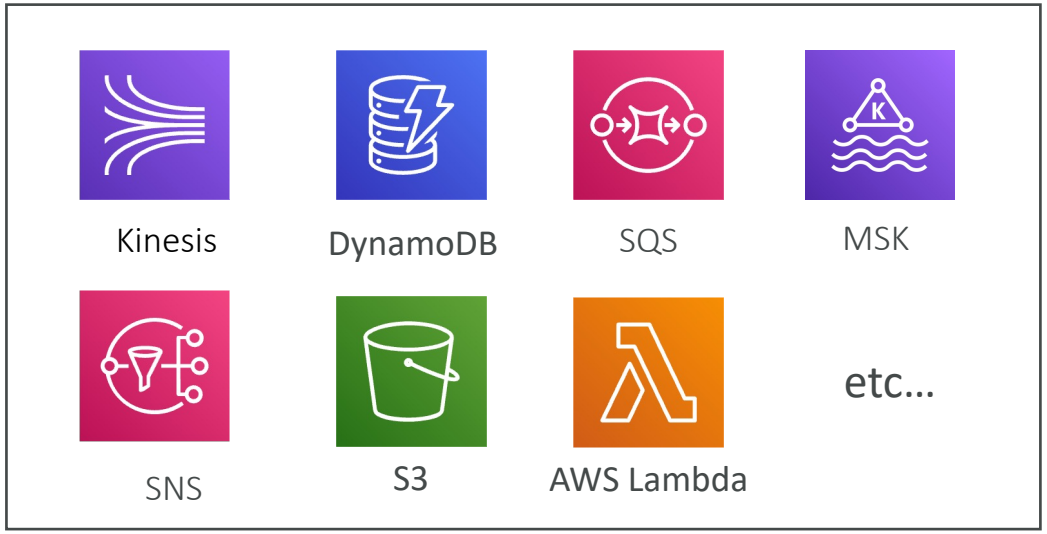


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IoT Core - Integrations

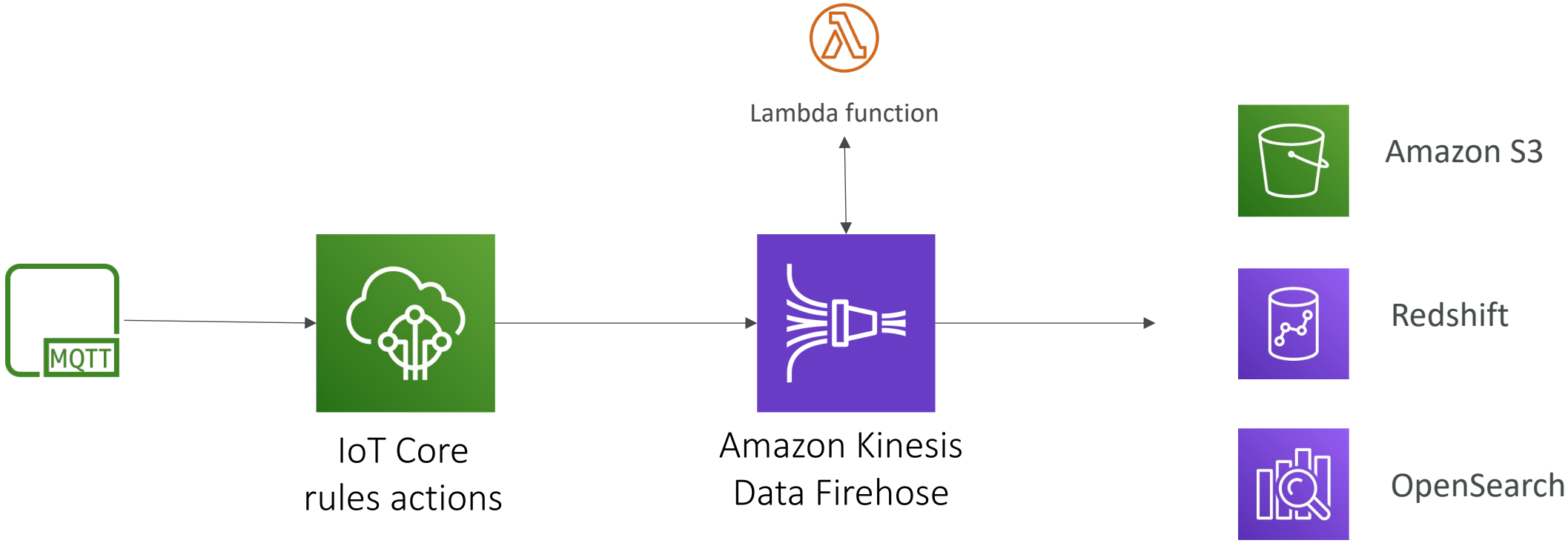




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IoT Core – Kinesis Data Firehose



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